SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I ***SHRUTHI K. (192571019*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SHRUTHI MUTHANNA K. (192571019)

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

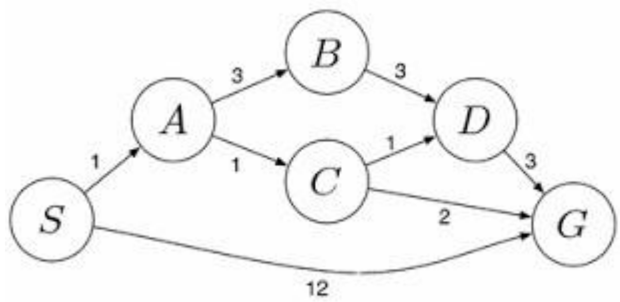
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ARTIFICIAL INTELLIGENCE – DETAILED ANSWERS

1. Water Jug Problem

The Water Jug problem is a classical Artificial Intelligence search problem. The objective is to obtain exactly two gallons of water in the 4-gallon jug using only the permitted operations such as filling, emptying and transferring water between the jugs. The solution is obtained by exploring valid states until the goal state is reached.

- Start with both jugs empty (4,0) and (3,0).
- Fill the 4-gallon jug completely.
- Pour water into the 3-gallon jug until it is full, leaving 1 gallon.
- Empty the 3-gallon jug.
- Transfer the remaining 1 gallon into the 3-gallon jug.
- Fill the 4-gallon jug again.
- Pour water into the 3-gallon jug until it becomes full. Exactly 2 gallons remain in the 4-gallon jug.
- Thus the goal state is achieved.

2. Mars Rover

The Mars Rover is an intelligent autonomous agent designed to explore Mars without continuous human control. It senses the environment, analyses information, plans actions and performs scientific tasks safely and efficiently.

- Percepts: cameras, sensors, rocks, terrain and weather.
- Environment: dynamic, partially observable and uncertain.
- Actions: move, avoid obstacles, collect samples, analyse rocks and transmit data.
- Performance: navigation accuracy, energy efficiency, communication reliability and successful mission completion.
- A model-based goal agent is most suitable because it maintains an internal model and makes intelligent decisions.

3. Eight Queens Problem

The Eight Queens problem is a constraint satisfaction problem in Artificial Intelligence. The aim is to place eight queens on a chessboard so that no queen attacks another queen horizontally, vertically or diagonally.

- Initial state: empty board.
- State: current arrangement of queens.
- Operators: place one queen safely in each column.
- Goal: all eight queens placed without conflict.
- Backtracking or DFS explores possible states until a valid arrangement is found.

4. OLA Cab Booking

Booking a cab through the OLA application can be modeled as a goal-based problem-solving agent. The system analyses user requirements and selects the most suitable cab automatically.

- Input pickup and destination.
- Find available cabs.
- Compare fare, distance and waiting time.
- Choose the best cab and confirm booking.
- This provides faster service and better customer satisfaction.

5. Uniform Cost Search

Uniform Cost Search (UCS) is an uninformed search algorithm that always expands the node with the lowest cumulative path cost. It guarantees the least-cost solution when path costs are non-negative.

- Insert source node into a priority queue.
- Expand the least-cost node.
- Update neighbouring path costs.
- Repeat until the goal node is reached.
- Applications include logistics, GPS navigation and route optimisation.